Devashree Gupta

dgupta16@jh.edu | 443-325-6025 | linkedin.com/in/devashreegupta | Baltimore, MD

Research Interests

Sensor characterization and calibration; analog and mixed-signal circuit design; embedded systems; signal processing for biomedical and RF applications; verification and validation of medical devices.

Education

Johns Hopkins University **Baltimore, MD**

B.S., Biomedical Engineering & B.S., Electrical Engineering (Dual Degree) *Aug. 2023 – May 2027 (Expected)*

Cumulative GPA: 3.68

Relevant Coursework: Control Systems, Nonlinear Dynamics, FPGA Programming, Microfabrication Lab, Biological Models and Simulations, Digital Signal Processing, VLSI, Advanced Electronics Design

Research Experience

Research Assistant, Engineering *July 2025 – June 2026*

Department of Biomedical Engineering, Johns Hopkins University *Baltimore, MD*

- Designed and built surveillance traps for mosquito attraction, capture, powder marking, and release.
- Designed a relay control PCB in KiCAD and developed an Arduino-based embedded system for automated nightly operation.
- Isolated repeated field failures of the trap visual system to insufficient illumination and re-designed lighting to improve image quality.

Signal Processing Engineering Intern *May 2025 – Aug. 2025*

Comillas Pontifical University *Madrid, Spain*

- Denoised raw satellite drone data and designed wavelet transforms to preprocess drone RF signatures.
- Developed a convolutional neural network achieving 88% accuracy classifying radio frequency signatures from drones.
- Applied FFT, wavelet transforms, and filtering to improve SNR by 40%, enhancing detection reliability.

Data Science Intern *July 2024 – Sept. 2024*

Brain Value Ltd. *Bangalore, India*

- Collected, processed, and analyzed biomarker and clinical trial datasets using machine learning methods; prepared reports and visualizations to characterize biomarker relationships.
- Processed clinical trial data, applying regression analysis, clustering, and predictive modeling using scikit-learn.

Industry Experience

Sensor Hardware Engineering Intern *June 2026 – Aug. 2026*

Medtronic *Lafayette, CO*

- Characterized voltage and current sensing across the full operating range of a next-generation surgical generator.
- Automated the full characterization loop in Python, sweeping loads, PWM drive modes, and harmonics on a self-built bench mount.
- Defined sensor gain and phase values from characterized data, delivering calibration baselines used in production firmware.
- Root-caused discrepancies between external probe and onboard measurements to unaccounted parasitic capacitance of traces.

Product Development Engineering Intern *Jan. 2026 – June 2026*

Johnson & Johnson MedTech *Baltimore, MD*

- Built an end-to-end cavitation test protocol for blood pump performance, enabling repeatable compliance testing across builds.
- Designed and built a high-speed camera setup to detect cavitation formation over the impeller inlet.
- Investigated de-airing methods for blood pumps to prevent gas embolism, identifying air localization across operating conditions to support next-generation design.

- Drafted and executed Test Method Validation protocols and feasibility studies of critical components supporting an IDE submission, utilizing Minitab and a Keyence IM.

Patents & Intellectual Property

Multi-Port Airway Adaptor — Provisional Patent Filed *Jan. 2025 – Present*

- Prototyping Lead for an airway adaptor enabling unobstructed visualization through multi-tool use during endotracheal intubation, supporting early detection of pulmonary hemorrhage by interventional pulmonologists.
- Modeled the port assembly in SolidWorks, designing clamping and sealing mechanisms; fabricated iterations via resin 3D printing with cast sealing elements.
- Conducted a needs assessment through review of 500+ clinical studies and operating-room shadowing of pulmonologists to analyze current device limitations.

Independent Research & Design Projects

Verification & Validation Lead — Transperineal Biopsy Collection System *Jan. 2024 – Aug. 2024*

- Led the design of the V&V plan for a new transperineal biopsy collection system to improve prostate cancer patient diagnosis.
- Designed and built test fixtures to characterize needle firing velocity using LVDT sensors, establishing the verification method for the mechanism's core design requirement.
- Completed a full working biopsy-gun prototype designed against FDA and ISO requirements; validated through a range of low-to-high fidelity testing methods.

CMOS Photosensor *Feb. 2025 – May 2025*

- Developed and implemented a 2.5 mm × 2.5 mm CMOS image sensor in Cadence Virtuoso, integrating 3-transistor active pixel sensors (APS) with photodiodes, row shift registers, and column scanning logic.
- Performed comprehensive DRC/LVS verification successfully.

Password-Encrypted Digital Keypad *Feb. 2025 – May 2025*

- Designed a 7-digit digital keypad system using Moore finite state machine architecture with one-hot encoding across 8 sequential passcode-validation states.
- Built a hardware prototype with D flip-flops, logic gates, and a clock circuit, using pull-down resistors to resolve current leakage and LED indicators for user feedback.

LED Music Sync Circuit *Oct. 2024 – Dec. 2024*

- Designed a circuit to synchronize LED lights to audio frequencies using an electret microphone, built on a breadboard to unit test amplification and filtering stages.
- Verified performance and optimized frequency response in LTSpice simulation.

BRAF V600E Mutation Study *Feb. 2024 – Apr. 2024*

- Analyzed BRAF V600E oncogene mutation impact on the MAP Kinase/ERK signaling pathway, producing cellular metrics with CellProfiler to evaluate its role in uncontrolled cell proliferation.
- Developed heatmaps, line graphs, and statistical analyses in MATLAB comparing oncogenic vs. wild-type populations, showing significantly higher ERK activity, more cytoplasmic localization, and smaller nuclei in oncogenic populations.

Teaching Experience

Physics Learning Assistant *Jan. 2025 – Dec. 2025*
Johns Hopkins University *Baltimore, MD*

- Facilitated group discussions, problem-solving, and reinforcement of concepts for electricity and magnetism courses, collaborating with professors and teaching assistants to improve student learning.

Teaching Assistant, First-Year Seminar: Design Cornerstone
Johns Hopkins University *Baltimore, MD*

First-Year Mentor *Aug. 2024 – Dec. 2025*
Johns Hopkins University *Baltimore, MD*

- Facilitated orientation activities and provided academic and campus resource support, hosting group events for incoming students.

Technical Skills

- **Programming & Data Analysis:** Python (NumPy, SciPy, pandas, scikit-learn), C/C++, MATLAB, LabVIEW, NI DAQ, SQL, Git, Minitab
- **Circuits & Embedded Systems:** KiCAD, LTSpice, Cadence Virtuoso, Verilog, Arduino
- **CAD & Prototyping:** SolidWorks (incl. FEA Simulation), Rapid Prototyping, 3D Printing (SLA/FDM), Machining (CNC, Lathe, Weld, Mill, Laser Cutting)

Leadership & Service

- **Professional Development Chair, AKDPhi (Asian-Interest Sorority):** Hosted resume workshops, networking events, and a headshot event.
- **Fundraising Chair, AKDPhi:** Raised $3,000 for breast cancer awareness through fundraising activities.
- **Hopkins Community Connection Advocate** (Jan. 2024 – May 2025): Assisted underserved patients and families at the Johns Hopkins Outpatient Center in overcoming barriers to social determinants of health.
- **Believe in Art Volunteer:** Volunteered at the Keswick Multi-Care Center, spending time and crafting with older patients living with dementia.